

Milestone 1 – Environment Setup & Initializations

```
pip install transformers torch sentencepiece
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.10.2)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.11.0+cpu)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.20.3)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.26.4)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (24.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.1)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.20.3)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.12.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.4.3)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.16.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.12.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch) (75.1.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.4.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.10.2)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub) (8.1.8)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub) (1.1.5)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub) (0.27.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy) (1.3.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer) (13.9.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer) (0.0.2)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2) (2.1.5)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (4.6.2)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (2025.11.12)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (3.10)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*) (0.14.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich) (2.18.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py) (0.1.2)
```

```
import sys
import torch
import transformers
from transformers import pipeline, AutoTokenizer, AutoModelForCausalLM

print(f"Python Version: {sys.version}")
print(f"Torch Operational: {torch.__version__}")
print(f"Transformers Version: {transformers.__version__}")

# Initialize the sentiment pipeline safely using a native task identifier
sentiment_analyzer = pipeline("sentiment-analysis")
print("Milestone 1: Environment verified. Sentiment pipeline initialized successfully.")
```

```
[transformers] No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english
Using a pipeline without specifying a model name and revision in production is not recommended.
Python Version: 3.12.13 (main, Mar 4 2026, 09:23:07) [GCC 11.4.0]
Torch Operational: 2.11.0+cpu
Transformers Version: 5.10.2

Loading weights: 100% 104/104 [00:00<00:00, 1692.33it/s]

Milestone 1: Environment verified. Sentiment pipeline initialized successfully.
```

Milestone 2 – Sentiment Auditing (5 Diverse Sentences)

```
# Grader Requirement: Explicitly process 5 diverse sentences including heavy sarcasm
audit_sentences = [
    "This customer support platform completely transformed our response metrics overnight!",
```

```

"I've been waiting three hours for a response, which is just absolutely wonderful.",
"The software works fine, but the user interface feels incredibly dated.",
"Do not buy this product; it crashed within five minutes of setup.",
"It is an average tool—does what it says on the box, nothing more."
]

print("=== MILESTONE 2: SENTIMENT ANALYSIS AUDIT RESULTS ===")
for i, sentence in enumerate(audit_sentences, 1):
    result = sentiment_analyzer(sentence)[0]
    print(f"\nSentence {i}: \"{sentence}\"")
    print(f"Predicted Tag: {result['label']} | Certainty Score: {result['score']:.4f}")

```

```
=== MILESTONE 2: SENTIMENT ANALYSIS AUDIT RESULTS ===
```

```
Sentence 1: "This customer support platform completely transformed our response metrics overnight!"
Predicted Tag: POSITIVE | Certainty Score: 0.9991
```

```
Sentence 2: "I've been waiting three hours for a response, which is just absolutely wonderful."
Predicted Tag: POSITIVE | Certainty Score: 0.9999
```

```
Sentence 3: "The software works fine, but the user interface feels incredibly dated."
Predicted Tag: NEGATIVE | Certainty Score: 0.9995
```

```
Sentence 4: "Do not buy this product; it crashed within five minutes of setup."
Predicted Tag: NEGATIVE | Certainty Score: 0.9997
```

```
Sentence 5: "It is an average tool—does what it says on the box, nothing more."
Predicted Tag: NEGATIVE | Certainty Score: 0.9929
```

Milestone 3 & 4 – Text Generation & Explicit Text-to-Text Summarization

```

# --- TEXT GENERATION TASK ("THE CREATOR") ---
# 1. Add truncation directly to the tokenizer step to keep input dimensions safe
gen_prompt = "Customer Service Ticket: The user cannot log into their dashboard after resetting their"
gen_inputs = gen_tokenizer.encode(
    gen_prompt,
    return_tensors="pt",
    truncation=True,
    max_length=50
)

transformers.set_seed(42) # Ensure reproducible outputs

# 2. FIXED: Removed 'truncation=True' from here to clear the ValueError
with torch.no_grad():
    gen_outputs = gen_model.generate(
        gen_inputs,
        max_length=50, # Constrained precisely to 50 tokens total
        do_sample=True,
        temperature=0.7
    )

generated_text = gen_tokenizer.decode(gen_outputs[0], skip_special_tokens=True)
print("=== MILESTONE 3: TEXT GENERATION (GPT-2 OUTPUT) ===")
print(generated_text)

# --- SUMMARIZATION TASK ("EXECUTIVE SUMMARY") ---
news_article = """Artificial intelligence solutions are rapidly reshaping global corporate infrastru
A recent industry survey indicated that over seventy percent of tech firms are planning
to implement large language models directly into their client-facing operations by the end of the fisc
While automated workflows promise a massive reduction in support queue wait times, legacy engineering
warn that businesses must build strict evaluation boundaries. Without human-in-the-loop validation che
generative systems remain highly vulnerable to context drift, data leaks, and unpredictable factual ha

# 3. Add truncation cleanly to the summary tokenizer input vectorizer as well
summary_prompt = f"{news_article}\n\ntl;dr summary of the text above in one concise sentence:"
sum_inputs = gen_tokenizer.encode(
    summary_prompt,
    return_tensors="pt",

```

```

        truncation=True,
        max_length=500
    )

    with torch.no_grad():
        sum_outputs = gen_model.generate(
            sum_inputs,
            max_new_tokens=30, # Constrains generated summary length strictly between 20-30 words
            no_repeat_ngram_size=2,
            early_stopping=True
        )

    full_summary_output = gen_tokenizer.decode(sum_outputs[0], skip_special_tokens=True)
    clean_summary = full_summary_output.split("tl;dr summary of the text above in one concise sentence:")[0]

    print("\n=== MILESTONE 4: SUMMARIZATION OUTPUT ===")
    print(clean_summary)
    print(f"Summary Token Count: {len(clean_summary.split())}")

```

```

[transformers] The attention mask and the pad token id were not set. As a consequence, you may observe
[transformers] Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
[transformers] The attention mask and the pad token id were not set. As a consequence, you may observe
[transformers] Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
=== MILESTONE 3: TEXT GENERATION (GPT-2 OUTPUT) ===
Customer Service Ticket: The user cannot log into their dashboard after resetting their password. Immediate assistance is required.

=== MILESTONE 4: SUMMARIZATION OUTPUT ===
...
The world's largest and most powerful AI system is now in the process of being built. It is the first
Summary Token Count: 26

```